**Learning Objective 4.4:** I can implement a PAM link that transmits and receives ASCII text end-to-end.

Show your work. Write a **Documentation** line at the end (**None** if you did not collaborate).

1. **Text to bits and back.** The L10 transmitter sends each byte most significant bit first. In ASCII, 'H' is 0x48 and 'i' is 0x69.

   (a) Write the 16 bits the transmitter sends for "Hi", in order.

   0x48 is 01001000 and 0x69 is 01101001, so the bits are `01001000 01101001`.

   | 01001000 01101001 |
   |---|

   (b) A receiver's Repack Bits block is set to LSB first by mistake. What byte does it produce from the first 8 bits, and does it print as a letter?

   Read in reverse, `01001000` becomes `00010010`, which is 0x12. That is a control code, not a printable character, so the line arrives as nonsense.

   | 0x12, no |
   |---|

   (c) A correctly set receiver outputs `01000111 01001111`. What two characters arrived? ('A' is 0x41, and the capital letters follow in order.)

   `01000111` is 0x47, six after 0x41, which is 'G'. `01001111` is 0x4F, fourteen after 0x41, which is 'O'. The text is "GO".

   | GO |
   |---|

2. **One packet.** A cadet changes the L10 message to `MAYDAY 41.ON 104.8W`, 19 characters. The Vector Source adds a newline, and the packet format is the L10 one: a 64-bit access code, a 32-bit header, then the payload. The link runs at 4 kbaud with one bit per symbol.

   (a) How many bits are in one packet, and how long does it take to send?

   The payload is $19 + 1 = 20$ bytes.

   $$n = 64 + 32 + 8 \times 20 = 256 \text{ bits}$$
   $$T = \frac{256}{4000 \text{ baud}} = 64 \text{ ms}$$

   bits: 256

   time: 64 ms

   (b) Write the 16-bit length field, which the header sends twice.

   The length is 20 bytes, which is 0000000000010100 in 16 bits.

   0000000000010100

   (c) What fraction of the bits on the air are text? If the same packets ran at 9.6 kbaud, the rate of L9's channel example, how long would one take?

   $$\frac{160}{256} = 62.5\%, \qquad \frac{256}{9600 \text{ baud}} = 26.7 \text{ ms}$$

   text: 62.5%

   time: 26.7 ms

3. **The access-code search.** Correlate Access Code - Tag Stream compares the last 64 bits with the access code after every new bit and declares a packet when at most 2 bits differ.

   (a) While no packet is arriving, the bits are random. About how many of the 64 match the access code at a typical position, and why?

   About 32 bits match. Each random bit matches the code's bit with probability one half, so on average half of the 64 match.

   (b) The link's bit error rate is $10^{-3}$. Find the fraction of packets the receiver misses because of errors in the access code with a threshold of 0, and with a threshold of 2.

   With a threshold of 0, one error anywhere in the 64 bits loses the packet:

   $$1 - (1 - 10^{-3})^{64} = 6.2\%$$

   With a threshold of 2, it takes 3 or more errors, and the 3-error term dominates:

   $$\binom{64}{3}(10^{-3})^3(1 - 10^{-3})^{61} = 3.9 \times 10^{-5}$$

   Including 4 or more errors gives $4.0 \times 10^{-5}$.

   threshold 0: $6.2\%$

   threshold 2: $4.0 \times 10^{-5}$

   (c) On the PlutoSDR version of the link, no packets arrive, yet the Symbol Sync output is clean and tightly clustered at $\pm 1$. Explain in words what the correlator sees and why, and what the receiver should change.

   The signal arrives inverted: the carrier-recovery loop has locked $180^\circ$ off, so every 1 arrives as a 0. Where the access code passes, every bit differs, so the correlator counts 0 of 64 matches instead of 64, and it never declares a packet. The fix is to flip the polarity, multiplying the samples by $-1$ before the slicer, which is what the lab flowgraph's polarity switch does.

4. **How much noise a line survives.** In the L10 link, the SNR at the slicer is $8/\sigma^2$ for a noise amplitude $\sigma$, each bit is wrong with probability $Q(\sqrt{\text{SNR}})$, and a 31-byte line arrives intact only if all 280 bits of its header and payload are right.

   (a) At $\sigma = 0.9$, find the SNR in dB, the bit error rate, and the fraction of lines that arrive intact. Use $Q(3.14) = 8.4 \times 10^{-4}$.

   $$\text{SNR} = \frac{8}{0.81} = 9.88 = 9.9 \text{ dB}$$
   $$\text{BER} = Q(\sqrt{9.88}) = Q(3.14) = 8.4 \times 10^{-4}$$
   $$P = (1 - 8.4 \times 10^{-4})^{280} = 0.79$$

   About 79% of the lines arrive intact; the lesson measured 76%.

   SNR: 9.9 dB

   lines: 79%

   (b) The link must deliver 99% of lines. Find the largest bit error rate that allows this, the SNR it needs (use $Q(3.97) = 3.6 \times 10^{-5}$), and the largest $\sigma$ the link tolerates.

   $$\text{BER} = 1 - 0.99^{1/280} = 3.6 \times 10^{-5}$$
   $$\text{SNR} = 3.97^2 = 15.8 = 12.0 \text{ dB}$$
   $$\sigma = \sqrt{8/15.8} = 0.71$$

   SNR: 12.0 dB

   $\sigma$: 0.71

   (c) A teammate proposes 200-byte packets to raise the share of text on the air from 72% to 94%. In words, explain why the link then needs more SNR for the same 99% of packets, and what a Lesson 2 link budget should do about it.

   The receiver loses a packet if any one of its bits is wrong, and a 200-byte packet has 1632 header and payload bits instead of 280, so each bit must be about six times more reliable for the packet to arrive as often. That takes about 12.8 dB instead of 12.0 dB. The link budget should target the SNR the longest packet needs, plus the usual fade margin.

5. **Clock error.** Two radios each derive their clocks from a crystal accurate to 20 ppm, so the receiver's sample clock can differ from the transmitter's symbol clock by up to 40 ppm. The link runs at 4 kbaud.

   (a) A receiver without timing recovery samples at fixed intervals. How many symbols, and how many seconds, until its sampling point slips one whole symbol? How many 344-bit packets is that?

   time: 6.25 s

   packets: 73

$$N = \frac{1}{40 \times 10^{-6}} = 25{,}000 \text{ symbols}$$

$$t = \frac{25{,}000}{4000} = 6.25 \text{ s}$$

$$\frac{25{,}000}{344} = 73 \text{ packets}$$

   (b) The L10 link loses almost no lines up to 5000 ppm with Symbol Sync in the receiver. Explain why the slip in part (a) does not appear in that link, and why a 40 ppm error is not a concern for it.

   Symbol Sync measures the timing error from the samples near each zero crossing and moves its sampling instant to follow the drift, so the sampling point never slips. 40 ppm is more than 100 times smaller than the 5000 ppm the link handles, so the loop follows it.

   (c) In the clock-error screenshot, the eye sink looks smeared while every line arrives. Explain why the eye sink and Symbol Sync disagree.

   The eye sink cuts the waveform every 8 samples, the nominal samples per symbol, so when each symbol spans slightly fewer samples, each trace starts a little later in its symbol than the last and the eye drifts across the display. Symbol Sync follows the actual symbol timing, so its samples stay at the eye's center.

**Documentation:** _____